N. Thirugnanamuthu's MB-Universal Matrix: Global Economic & Tech Transition Model

N. Thirugnanamuthu
Mechanical Engineer, Data Scientist & Data Analyst
Independent Researcher

Abstract

We present the MB-Universal Matrix, a matrix framework for modelling how global economic, technological, social and environmental indicators evolve and respond to external actors. The framework has three parts: a column-stochastic transition matrix A, a third-order impact tensor $\mathscr{T}$ that perturbs A, and a stochastic simulation layer. We prove that $\rho(A) = 1$, give a spectral convergence bound, and derive the ordinary and ridge least-squares estimators step by step. As a first empirical test, a ridge VAR(1) is fitted to five World Bank world-aggregate indicators for 2005–2023 ($n = 17$ transitions). Rolling-origin validation shows no forecasting skill over a mean baseline (overall skill $+0.001$), the fitted spectral radius is 0.094 (90% bootstrap interval 0.049–0.204), and the headline GDP lag-1 coefficient $b = -0.516$ ($t = -2.33$) falls to $b = -0.273$ ($t = -1.11$) when three COVID-affected pairs are removed. We therefore report the framework as a validated mathematical model and a* negative *first empirical result: world-aggregate annual data are too short to identify cross-indicator dynamics. A panel extension is proposed.

Index Terms—Transition matrix, Markov chain, Perron–Frobenius, ridge regression, vector autoregression, bootstrap, World Bank data.

I. Introduction

Economic growth, digital diffusion, research effort, emissions and capital flows interact over time. A compact way to describe such interaction is a linear transition law $X_{t+1} = AX_t$ in which the entries of A encode how each indicator feeds the others. This paper defines such a law, equips it with an impact tensor for external actors, proves its basic properties, and then asks the empirical question: *can A be estimated from real world-level data?*

The contributions are: (i) the MB-Universal Matrix definition (Section III); (ii) hand-checkable proofs of the spectral properties and of the estimators (Section IV); (iii) an honest empirical test on World Bank data with out-of-sample validation (Section V).

II. Data

All empirical series are the World aggregate (code WLD) from the World Bank Open Data API [5], 2005–2023 ($T = 19$ years): GDP growth (annual %, NY.GDP.MKTP.KD.ZG), individuals using the Internet (% of population, IT.NET.USER.ZS), R&D expenditure (% of GDP, GB.XPD.RSDV.GD.ZS), CO_2 emissions per capita (t, EN.GHG.CO2.PC.CE.AR5) and net FDI inflows (% of GDP, BX.KLT.DINV.WD.GD.ZS). 2005 is the first year in which the Internet series is available. Fig. 1 shows the standardised levels; the shaded band marks 2020.

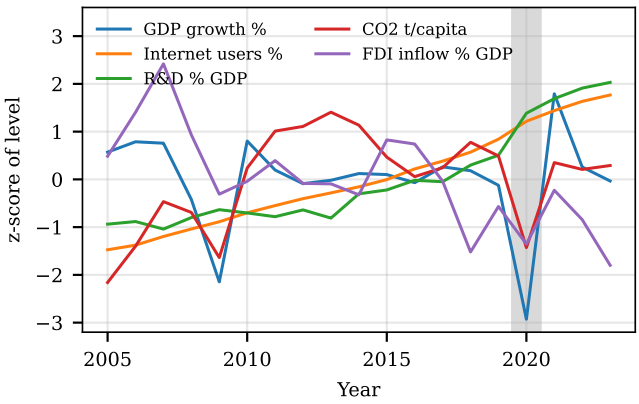


Figure 1. World indicators, z-scored levels, 2005–2023 (World Bank).

III. The MB-Universal Matrix

A. State vector and transition matrix

Let $X_t \in \Delta^{n-1} = \{X \geq 0,\ \mathbf{1}^\top X = 1\}$ be the normalised state of $n = 5$ components: decision power, economy, AI growth, human mindset, environment.

Definition 1. $A \in \mathbb{R}^{n\times n}_{>0}$ is *column-stochastic* if $\mathbf{1}^\top A = \mathbf{1}^\top$. The model is $X_{t+1} = AX_t$.

The assumed (not estimated) matrix of the framework is

$$A = \begin{bmatrix} 0.70 & 0.10 & 0.15 & 0.05 & 0.02 \\ 0.15 & 0.65 & 0.10 & 0.10 & 0.02 \\ 0.05 & 0.10 & 0.65 & 0.05 & 0.10 \\ 0.05 & 0.10 & 0.05 & 0.75 & 0.06 \\ 0.05 & 0.05 & 0.05 & 0.05 & 0.80 \end{bmatrix} \tag{1}$$

(after column normalisation, each column sums to 1).

B. Impact tensor

Let $\mathscr{T} \in \mathbb{R}^{n\times n\times m}_{\geq 0}$, with slice $T_k = \mathscr{T}_{:,:,k}$ the influence matrix of actor k ($m = 3$: technology leader, investor, social activist).

With intensities $u \in \mathbb{R}^m_{\geq 0}$,

$$A_{\text{eff}}(u) = \text{cn}\Big(A + \sum_{k=1}^{m} u_k T_k\Big), \quad X_{t+1} = A_{\text{eff}}(u) X_t + \sigma \varepsilon_t, \quad (2)$$

where $\text{cn}(\cdot)$ normalises columns to sum 1 and ε_t is Gaussian noise followed by projection onto the simplex.

C. Empirical (estimated) form

For real data the state is the vector of standardised annual changes $z_t \in \mathbb{R}^5$ and the model is the ridge VAR(1) [1]

$$z_{t+1} = B z_t + e_t. \quad (3)$$

B is *not* stochastic; it is a regression map, and stability requires $\rho(B) < 1$.

IV. Step-by-Step Proofs

A. Spectral radius of A

Theorem 1. *If A is column-stochastic then $\rho(A) = 1$.*

Proof. Step 1. $\mathbf{1}^\top A = \mathbf{1}^\top$, so $A^\top \mathbf{1} = \mathbf{1}$: 1 is an eigenvalue of $A^\top$, hence of A (same spectrum). Thus $\rho(A) \geq 1$.
Step 2. The induced 1-norm is the maximum absolute column sum: $\|A\|_1 = \max_j \sum_i |a_{ij}| = 1$.
Step 3. For any eigenpair $Av = \lambda v$, $|\lambda| \|v\|_1 = \|Av\|_1 \leq \|A\|_1 \|v\|_1 = \|v\|_1$, so $|\lambda| \leq 1$.
Step 4. Combining Steps 1 and 3 gives $\rho(A) = 1$. Numerically, the eigenvalue moduli of Eq. (1) are $1, 0.750, 0.689, 0.557, 0.557$. ∎

B. Convergence to the stationary distribution

Since $A > 0$, Perron–Frobenius theory [2] gives a simple eigenvalue 1 with a positive right eigenvector π, $\mathbf{1}^\top \pi = 1$, and all other eigenvalues satisfy $|\lambda_j| < 1$. If A is diagonalisable, $X_t - \pi = \sum_{j \geq 2} c_j \lambda_j^t v_j$, hence

$$\|X_t - \pi\|_1 \leq C |\lambda_2|^t. \quad (4)$$

For Eq. (1): $\pi = (0.206, 0.210, 0.176, 0.208, 0.200)$, $|\lambda_2| = 0.75$, spectral gap 0.25. The time to shrink the error by a factor 100 is $t^* = \ln(0.01)/\ln(0.75) \approx 16$ steps. In the simulation the inequality holds with $C = \|X_0 - \pi\|_1$ at every step (Fig. 2b); this is an observation for this instance, not a general theorem.

C. Least-squares estimator (one variable)

Let (x_i, y_i), $i = 1, \ldots, N$, be the pairs (z_t, z_{t+1}) and $S(b) = \sum (y_i - b x_i)^2$.

$$\frac{dS}{db} = -2 \sum x_i (y_i - b x_i) = 0 \quad (5)$$

$$\Rightarrow \hat{b}_{\text{OLS}} = \frac{\sum x_i y_i}{\sum x_i^2}. \quad (6)$$

Because $\frac{d^2 S}{db^2} = 2 \sum x_i^2 > 0$ this is the minimum. With ridge penalty λb^2:

$$\frac{d}{db}\left[S(b) + \lambda b^2\right] = -2 \sum x_i y_i + 2b \sum x_i^2 + 2\lambda b = 0 \quad (7)$$

$$\Rightarrow \hat{b}_{\text{ridge}} = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda}. \quad (8)$$

Table 1. The 17 pairs $(x, y) = (z_t, z_{t+1})$ for GDP growth changes.

#	x	y	x^2	xy	#	x	y	x^2	xy
1	0.151	0.003	0.023	0.000	10	0.008	−0.083	0.000	−0.001
2	0.003	−0.693	0.000	−0.002	11	−0.083	0.221	0.007	−0.018
3	−0.693	−1.024	0.480	0.710	12	0.221	−0.028	0.049	−0.006
4	−1.024	1.805	1.049	−1.849	13	−0.028	−0.165	0.001	0.005
5	1.805	−0.348	3.260	−0.628	14	−0.165	−1.677	0.027	0.276
6	−0.348	−0.153	0.121	0.053	15	−1.677	2.878	2.813	−4.827
7	−0.153	0.064	0.023	−0.010	16	2.878	−0.908	8.283	−2.614
8	0.064	0.105	0.004	0.007	17	−0.908	−0.157	0.825	0.142
9	0.105	0.008	0.011	0.001					
			$\sum x^2 = 16.975$,					$\sum xy = -8.760$	

Table 2. Actor influence in the toy model.

Actor	$\|T_k\|_F$	$\|\Delta\pi\|_1/\Delta u$
Technology leader	1.360	0.658
Investor	1.021	0.386
Social activist	1.040	0.685

The shrinkage factor is $\sum x_i^2 / (\sum x_i^2 + \lambda) < 1$ [3]. For the multivariate case the same derivation gives $\hat{B}^\top = (Z_0^\top Z_0 + \lambda I)^{-1} Z_0^\top Z_1$.

D. Worked numerical example: GDP growth

Step 1 (changes). $d_t = g_{t+1} - g_t$, giving 18 values, e.g. $d_1 = 4.489 - 4.059 = 0.430$; $d_{15} = -2.886 - 2.680 = -5.567$ (2020); $d_{16} = 6.485 - (-2.886) = 9.371$ (2021).
Step 2 (moments). $\bar{d} = -0.0667$, $\sum (d - \bar{d})^2 = 182.811$, $s = \sqrt{182.811/17} = 3.279$.
Step 3 (z-scores). $z = (d - \bar{d})/s$, e.g. $z_{15} = -1.677$, $z_{16} = 2.878$.
Step 4 (products). Table 1 lists the 17 pairs. Totals: $\sum x^2 = 16.975$, $\sum xy = -8.760$.
Step 5 (OLS). $\hat{b} = -8.760/16.975 = -0.516$.
Step 6 (t-test). $\text{se} = \sqrt{\sum (y - \hat{b}x)^2/(N-2) / \sum x^2} = 0.2212$, $t = -0.516/0.2212 = -2.33$.
Step 7 (ridge, $\lambda = 100$). $\hat{b} = -8.760/(16.975 + 100) = -0.0749$.
Step 8 (source of the effect). Pairs 15 and 16 contribute $-4.827 - 2.614 = -7.441$, i.e. 85% of $\sum xy$.
Step 9 (COVID pairs 14–16 removed). $N = 14$, $\sum x^2 = 5.852$, $\sum xy = -1.595$, $\hat{b} = -0.273$, $\text{se} = 0.2465$, $t = -1.11$; ridge gives -0.0151.

V. Results

A. Toy model (assumed parameters)

Fig. 2 shows the stochastic MB-model with actor intensities $u = (0.15, 0.05, 0.02)$. The stationary AI share moves from 0.176 (no actors) to 0.155 (with actors). Over 300 Monte Carlo runs ($\sigma = 0.01$) the final shares are 0.197 ± 0.012 (decision), 0.221 ± 0.010 (economy), 0.156 ± 0.010 (AI), 0.225 ± 0.012 (mindset), 0.201 ± 0.013 (environment). Actor influence depends on the metric (Table 2): the technology leader has the largest tensor norm, the social activist the largest effect on the stationary distribution. These are properties of the *assumed* $\mathscr{T}$, not measurements of real persons.

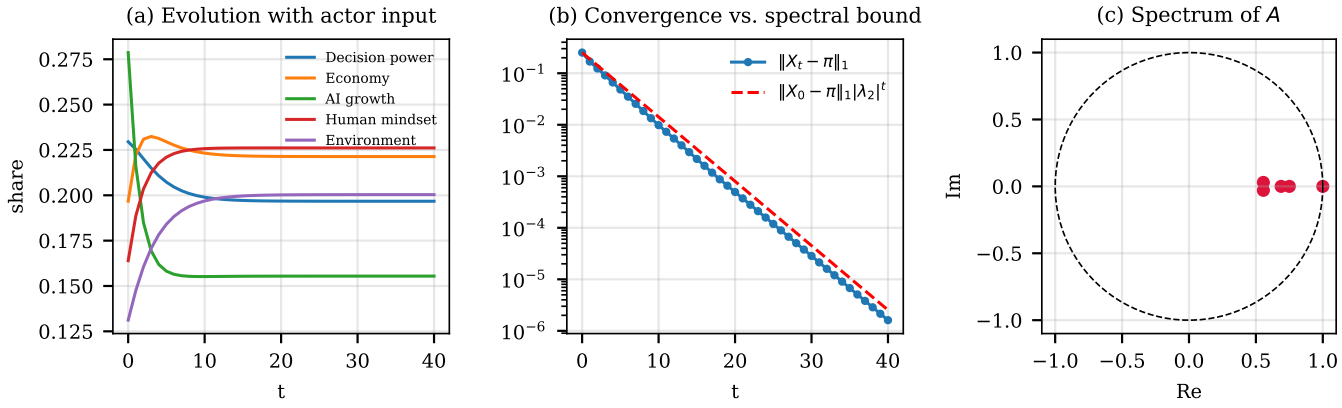


Figure 2. Toy MB-Universal Matrix: (a) evolution with actor input, (b) convergence compared with $|\lambda_2|^t$, (c) eigenvalues of A on the unit circle.

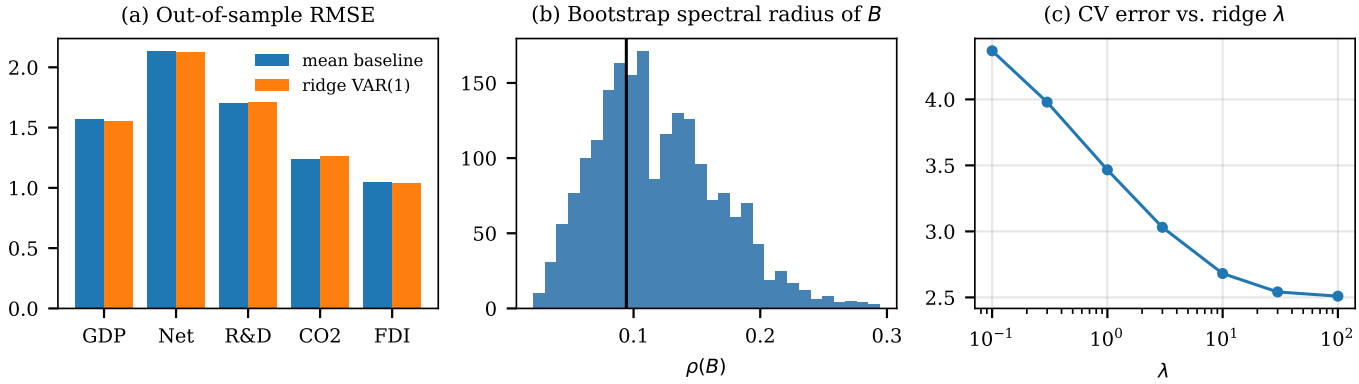


Figure 3. Validation: (a) out-of-sample RMSE versus mean baseline, (b) bootstrap distribution of $\rho(\hat{B})$, (c) cross-validation error versus ridge strength.

B. Empirical estimation

The ridge strength was chosen by rolling-origin cross-validation (train on the first s changes, predict change $s+1$, $s = 8, \ldots, 17$). The CV error decreased monotonically up to the largest value tried ($\lambda = 100$; Fig. 3c), i.e. the data prefer to shrink B almost to zero. The fitted matrix (rows: affected, columns: driver in the order GDP, Internet, R&D, CO_2, FDI) is

$$\hat{B} = \begin{bmatrix} -.059 & .034 & .056 & -.073 & -.047 \\ -.002 & .063 & .035 & -.025 & .024 \\ -.007 & .058 & -.002 & -.020 & .028 \\ -.022 & .011 & .017 & -.020 & -.036 \\ -.005 & -.005 & .066 & -.018 & -.013 \end{bmatrix}. \quad (9)$$

Its spectral radius is $\rho(\hat{B}) = 0.094$, with a 90% pairs-bootstrap interval [4] of 0.049–0.204 (2000 resamples). A year-on-year change therefore does not persist.

Table 3 compares one-step out-of-sample RMSE (z-units) with a mean baseline. The overall skill, $1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{base}}$, is $+0.001$, which is no improvement.

Five of the 25 entries have a 90% bootstrap interval that excludes zero (e.g. R&D→FDI $+0.07$). With 25 tests, about 2.5 such entries are expected by chance, and the model has no out-of-sample skill, so we do not interpret them as causal links.

Table 3. One-step out-of-sample RMSE, rolling origin.

Indicator	Model	Baseline	Skill
GDP growth	1.557	1.573	+0.02
Internet users	2.125	2.132	+0.01
R&D	1.709	1.703	−0.01
CO_2 per capita	1.266	1.241	−0.04
FDI inflow	1.042	1.047	+0.01

VI. Limitations

(1) $T = 19$ years gives 17 transitions for 25 parameters; the problem is under-determined without strong shrinkage. (2) The indicators are world aggregates; "decision power" and "human mindset" have no standard global measurement and were replaced by measurable proxies. (3) 2020 is an outlier (Fig. 4). (4) The toy parameters in Eq. (1) and $\mathscr{T}$ are assumptions. (5) The bootstrap resamples pairs and ignores serial dependence. (6) The framework models indicators, not named individuals.

VII. Conclusion

The MB-Universal Matrix is mathematically well defined: $\rho(A) = 1$, convergence is geometric at rate $|\lambda_2|$, and the esti-

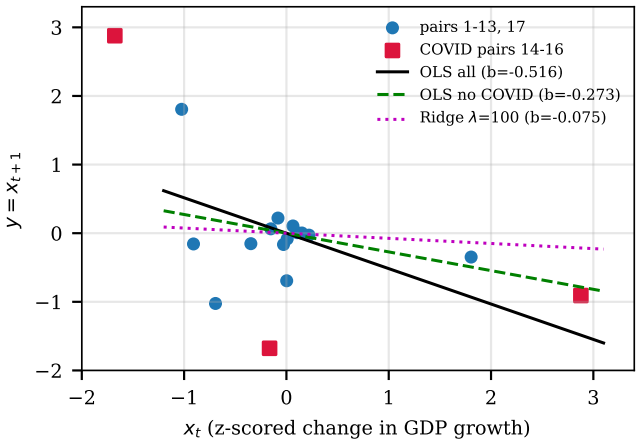


Figure 4. GDP lag-1 regression. Three COVID-affected pairs (squares) drive the negative slope.

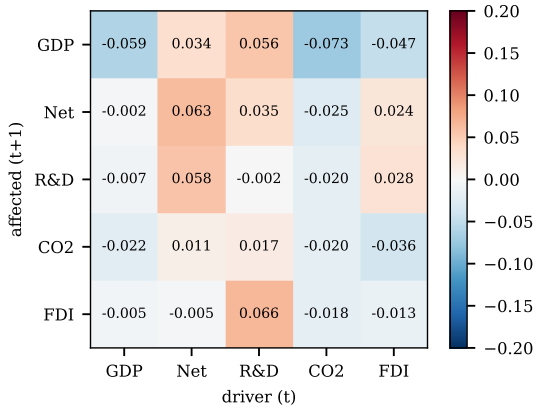


Figure 5. Estimated $\hat{B}$ (ridge, $\lambda = 100$).

mators were derived and verified by hand and by computation. On 2005–2023 world data, the estimated transition map has no forecasting skill over a mean baseline, and the apparent GDP lag-1 effect depends on three COVID-affected pairs. The natural extension is a country×year panel (about 150 countries) with country fixed effects, which raises the effective sample size by two orders of magnitude, together with longer indicator histories.

Data and Code Availability

The series are in `world_data_2005_2023.csv`; the analysis scripts are `universal_matrix_advanced.py` and `estimate_A_real_data.py`. Data source: World Bank, World Development Indicators, `https://data.worldbank.org`.

References

[1] H. Lütkepohl, *New Introduction to Multiple Time Series Analysis*. Berlin: Springer, 2005.

[2] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge Univ. Press, 2013.

[3] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[4] B. Efron, "Bootstrap methods: Another look at the jackknife," *Ann. Statist.*, vol. 7, no. 1, pp. 1–26, 1979.

[5] World Bank, "World Development Indicators," accessed Oct. 2026. [Online]. Available: https://data.worldbank.org